

Phase 2 Insight Report

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Executive Summary

Phase 2 of the Data Vortex challenge focused on rebuilding the Social Engine's analytical core using the cleaned social-media dataset. The dataset was structured into relational users and posts tables in PostgreSQL, with `user_id` establishing the relationship between users and their posts.

Three analytical SQL challenges were performed across different levels of difficulty. The Easy-level analysis identified the 10 most engaged posts using likes, shares, and comments. The Medium-level analysis compared average post engagement across platforms for users with at least 30,000 followers. The Hard-level analysis identified low-follower users whose total engagement falls within the top 10% of all users, highlighting potentially unusual high-performing or suspicious accounts.

The analyses demonstrate how relational joins, aggregation, filtering, CTEs, and window functions can be used to move from raw social activity data to meaningful behavioural and anomaly-based insights. Missing values were handled without fabricating data, ensuring that the resulting analysis remained based on observed values.

Overall, Phase 2 transforms the restored dataset into a structured analytical system capable of identifying engagement leaders, platform-level behaviour, and potential follower-to-engagement anomalies.

Schema Design & Table Structuring

The cleaned social-media dataset was structured into two relational tables in PostgreSQL: users and posts. The users table stores user-level attributes such as location, language, account creation date, and follower count, while the posts table stores post-level information including platform, timestamp, likes, shares, and comments.

The two tables are connected through **user_id**, which acts as the primary key in the users table and a foreign key in the posts table. This structure avoids unnecessary duplication of user information and allows user-level and post-level attributes to be analysed together.

Schema Structure:

Table	Purpose	Key fields
users	Stores user-level information	user_id, follower_count, location, language
posts	Stores individual post activity	post_id, user_id, platform, likes, shares, comments, timestamp

Relationship:

users.user_id → posts.user_id (one user can have multiple posts)

The database was validated after loading the cleaned data. The posts table contained 12,000 records, and a referential-integrity check returned 0 unmatched user IDs, confirming that every post was successfully linked to a valid user.

For missing values, the analysis retained the distinction between **missing and zero** rather than fabricating values. In particular, posts with missing likes were excluded from engagement calculations where actual likes were required.

E2 — Most Engaged Posts

Objective:

Identify the top 10 posts by total engagement, where total engagement is calculated as the sum of likes, shares, and comments. Posts with missing likes were excluded from the calculation.

Data Output

Messages

Notifications

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Key Findings:

- The highest-engagement post recorded 7,893 total engagements, consisting of 4,983 likes, 1,919 shares, and 991 comments.
- The remaining top-ranked posts also show consistently high engagement, with the 10th-ranked post recording 7,610 total engagements.
- The difference between the highest and 10th-highest posts is only 283 engagements, indicating a relatively close competition among the most-engaged posts.
- Likes form the largest component of total engagement for all ten posts, while shares and comments provide substantial additional interaction.
- The top 10 posts span multiple platforms, including Instagram, Reddit, Facebook, YouTube, and Twitter, rather than being concentrated on a single platform.
- Instagram appears multiple times among the top-performing posts, including the highest-engagement post.
- One of the top 10 posts has a missing platform value, but it remains in the result because platform was not a filtering condition for this challenge.
- Overall, the results identify a small group of posts generating exceptionally high interaction and provide a basis for further investigation into the characteristics of highly engaging content.

Insight:

The top-engagement results show that high-performing posts are distributed across multiple platforms, suggesting that strong engagement is not exclusive to one platform. The relatively narrow engagement range within the top 10 also indicates that several posts perform at a similarly high level rather than one post being an extreme outlier.

M4 — Platform Behaviour by High Follower Users

Objective:

Determine which platform provides the highest average engagement per post among users with at least 30,000 followers.

Data Output				Messages	Notifications
	platform character varying (20)	post_count bigint	avg_engagement_per_post numeric		
1	Instagram	647	4144.59		
2	Facebook	647	3990.95		
3	Reddit	650	3985.53		
4	Twitter	629	3977.66		
5	YouTube	683	3961.46		

Key Findings:

- Instagram records the highest average engagement per post at 4,144.59, making it the top-performing platform for high-follower users.
- Facebook ranks second with an average engagement of 3,990.95 per post.
- Reddit follows closely with 3,985.53 average engagement per post.
- Twitter records an average engagement of 3,977.66 per post.
- YouTube has the lowest average among the five platforms at 3,961.46 per post.
- The difference between Instagram and YouTube is 183.13 engagements per post, showing that the platform averages are relatively close.
- The number of qualifying posts is also fairly balanced across platforms, ranging from 629 posts on Twitter to 683 posts on YouTube.
- Since the analysis considers users with 30,000 or more followers, the results specifically represent platform behaviour among relatively high-follower accounts rather than the entire user base.

Insight:

Instagram emerges as the strongest platform for average engagement among high-follower users. However, the relatively small gap between the platforms indicates that high-follower accounts receive substantial engagement across all five platforms, rather than Instagram having a dramatically higher engagement level.

H4 — Follower to Engagement Anomaly

Objective:

Identify users with fewer than 5,000 followers whose total post engagement places them within the top 10% of all users, highlighting potentially unusual high-performing or suspicious accounts.

Data Output		Messages	Notifications		
					
			SQL	Show	
	user_id [PK] character varying (20)	follower_count integer	total_engagement numeric	engagement_percentile numeric	
1	user_fgjkkrie	2211	58480.0	0.0060	
2	user_hdas0iau	1620	55118.0	0.0107	
3	user_uerv85na	1824	52876.0	0.0153	
4	user_r7eg1rac	2052	50266.0	0.0240	
5	user_67hyf45u	898	47627.0	0.0367	
6	user_5oe5t3js	3151	47481.0	0.0380	
7	user_6wra58f7	4953	46435.0	0.0433	
8	user_n0ok02rt	2531	43894.0	0.0640	
9	user_06v0exkg	552	43669.0	0.0667	
10	user_txrypced	2479	42896.0	0.0747	
11	user_cqaitbuv	3382	42248.0	0.0820	
12	user_ha9bzru4	608	41826.0	0.0880	
13	user_knjtsbf3	1298	41804.0	0.0893	
14	user_rr1uzkql	1069	41762.0	0.0900	
15	user_lfwnie3f	2127	41707.0	0.0907	
16	user_0aacyvtz	1452	41499.0	0.0947	
17	user_zozhwbis	4580	41284.0	0.0967	

Key Findings:

- The analysis identified 17 users satisfying both conditions: fewer than 5,000 followers and an engagement percentile within the top 10%.
- The highest-engagement user in this group has only 2,211 followers but a total engagement of 58,480.
- The second-highest identified user has 1,620 followers and 55,118 total engagement.
- Several users achieve more than 50,000 total engagements despite having fewer than 5,000 followers, showing a strong follower-to-engagement imbalance.
- The identified users have follower counts ranging from 552 to 4,953, demonstrating that the pattern is present across different levels of low follower counts.
- The engagement percentile ranges from 0.0060 to 0.0967, placing every identified user within the top 10% engagement distribution.
- The user with only 552 followers records 43,669 total engagement, making the follower-to-engagement difference particularly notable.
- The lowest-engagement user among the identified accounts still records 41,284 total engagement with only 4,580 followers.
- These results do not by themselves establish that an account is suspicious; they identify accounts whose engagement performance is unusually high relative to their follower base and therefore warrant further investigation.

Insight:

The H4 analysis reveals a clear group of low-follower, high-engagement accounts. Some users generate engagement levels comparable to highly active accounts despite having relatively small follower bases. This may indicate unusually effective content performance or potentially anomalous behaviour, making these accounts suitable candidates for deeper investigation.

Conclusion

Phase 2 successfully rebuilt the Social Engine's analytical core using PostgreSQL and structured SQL analysis. The three challenges demonstrated different levels of analytical reasoning, from ranking highly engaged posts to comparing platform behaviour and identifying unusual follower-to-engagement patterns.

The analysis shows that SQL joins, aggregations, CTEs, and window functions can transform the cleaned dataset into actionable behavioural insights. While the H4 results highlight accounts that warrant further investigation, the identified patterns should not be treated as proof of suspicious activity without additional analysis.

Overall, the Phase 2 pipeline provides a reproducible foundation for trend detection, behavioural analysis, and anomaly discovery while maintaining data-driven and non-fabricated results.